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Due 1/20/26

Indicate the following languages corresponding to the following problems:

1. Given: a number *n* and two primes *p* and *q* is is it the case that *n* = *p* * *q

$$L = \{\langle n, p, q\rangle \mid p \text{ and } q \text{ are prime and } n = p \cdot q\}$$

2. Given a number *n* . is it the case *n* = *p* q *for some primes* q *and* q*?

$$L = \{\langle n\rangle \mid \exists \text{ primes } p, q \text{ such that } n = p \cdot q\}$$

3. Given an NFA *A* and a word *w* does A accept *w*?

$$L = \{\langle A, w\rangle \mid A \text{ is an NFA and } A \text{ accepts } w\}$$

4. Given an NFA *A* Is there a word *w* such that *A* accepts *w*?

$$L = \{\langle A\rangle \mid A \text{ is an NFA and } \exists w \text{ such that } A \text{ accepts } w\}$$

Prove why the following statements are true:

1. Function $2n^3$ - 18n is $O(n^3)$ and also it is $O(n^4)$ but is not $O(n^2 \log n)$

PF: c = 2, g(f) = $O(n^3)$

$$\frac{2n^3 - 18n}{2n^3} = 1 - \frac{9}{n^2} \xrightarrow[n\to\infty]{} 1 \leq 1, \quad \Rightarrow \quad 2n^3 - 18n \in O(n^3).$$

PF: c = 1, g(f) = $O(n^4)$

$$\frac{2n^3 - 18n}{n^4} = \frac{2}{n} - \frac{18}{n^3} \xrightarrow[n\to\infty]{} 0$$

PF: c = 1, g(f) = $O(n^2 \log n)$

$$\frac{2n^3 - 18n}{n^2 \log n} = \frac{2n}{\log n} - \frac{18}{n \log n} \xrightarrow[n\to\infty]{} \infty > 1, \quad \Rightarrow \quad 2n^3 - 18n \notin O(n^2 \log n).$$

2. Function $3n^2 2^{2n}$ is $2^{O(n)}$

PF: c = 3, g(f) = $2^{O(n)}$

$$\frac{3n^2 2^{2n}}{2^{3n}} = 3n^2 \cdot 2^{-n} = \frac{3n^2}{2^n} \xrightarrow[n\to\infty]{} 0 < 1, \quad \Rightarrow \quad 3n^2 2^{2n} \in 2^{O(n)}.$$

Proteins 3D shapes decide how they work. Better testing and predicating of protein structures from their amino acid sequences allow for faster drug discovery, understanding of diseases, and the design of new proteins for medicine and biotechnology. An algorithm *M* can measure the similarity between two proteins, helping scientists identify proteins with similar structures, predict their functions, and find or design drugs more efficiently.

To represent a protein in a computer. we would use a Graph. Each atom is a node and the bonds connecting each atom is a edge.
Pros: Very accurate and reflects the molecules structure
Cons: Very complex and can be slow to process for large proteins.

Comparing two similarity metrics between two protein molecules would be the sequence and the structure
Sequence: the order of amino acids in a protein represented as a string of letters
Pros: Easy to store and compare
Cons: does not use 3D shape
Structure: The distance in a graph (nodes = atoms, edges = bonds)
Pros: Very accurate comparison
Cons: Computationally heavy

Kuhlman, B., Bradley, P. Advances in protein structure prediction and design. *Nat Rev Mol Cell Biol* **20**, 681–697 (2019). https://doi.org/10.1038/s41580-019-0163-x